Naman Dixit

Jaipur, India

Google Scholar | GitHub | LinkedIn | sci.dixit@gmail.com

+91 75976 30305

RESEARCH INTERESTS

• Developing computational inverse-imaging frameworks at the intersection of Physics-Informed ML and non-equilibrium stochastic mathematical physics for open neurobiological systems, using Stochastic SINDy, Neural Operators, Mori–Zwanzig kernels, and Neural SDEs.

• Quantifying uncertainty in ultrafast nanoscale transport, decoherence, and non-Markovian memory across ion-channel dynamics, proton/electron tunneling, and mitochondrial electron transport.

EXPERIENCE

Tsinghua University | *MAIC Project* **Virtual**

Learning Experience Ambassador Sep 2026 – Oct 2026

• Modeled AI-induced cognitive offloading using *CNFs*, *Neural SDEs*, and *Volterra kernels* to capture measurement ambiguity, state transitions, and feedback dynamics.

• Demonstrated **24.9**% boundary contraction across **4,500**+ simulations and reduced held-out EEG MSE by **25.56**% on OpenNeuro using a drift-feedback-memory model.

National Science Foundation (*NSF*) | *IAIFI Summer School* **Virtual**

Participant Aug 2026

• Engineered *InverseBench* pipelines for plug-and-play diffusion-prior benchmarking and formulated a PIML inverse problem for magnetization recovery and spin coherence, informed by Prof. Berthy Feng's insights.

KIAS-KAIST | *7th Electronic Structure Calculations Summer School* **Virtual**

Participant Jul 2026

• Trained in **high-performance computing** parallelization and numerical execution of **master-equation models** during the summer school.

• Examined stochastic master-equation models for open-system dynamics and investigated transport and electronic-structure methods, including *ROM*, *TDDFT*, and *RTA*.

University of California, Berkeley | *REYES Programme* **Virtual**

Research Scholar Jun 2026 – Jul 2026

• Investigated **Wigner functions** and **symmetry structures** through *STEM-H* training, applying these frameworks to quantum dynamics, molecular biophysics, and complex-manifold modeling.

PROJECTS

Lindblad Neural Operators (LNO) for Nanoscale Imaging GitHub

• Engineered the Lindblad Neural Operator *(LNO)* framework by embedding learnable jump operators and geometric matrix projections into **Fourier spectral kernels** to enforce continuous semigroup properties and positive-semidefinite (**PSD**) constraints.

• Eliminated the baseline FNO's **34.08**% nonphysical boundary-violation rate, achieving *zero PSD* violations across **100**-step rollouts; under OOD noise, achieved a relative L_2 error of **0.100** versus **0.155** for FNO.

PUBLICATIONS

Stochastic (PNP) Dataset for Neurobiological Ion Transport | *Harvard Dataverse* Dataset

Dixit, N. | *DOI: 10.7910/DVN/IUMEDG* Jul 2026

• Published an open-access dataset of ~**3,000** stochastic Poisson–Nernst–Planck (*PNP*) simulations for non-equilibrium ion transport, with reproducible train/validation/test splits and a FAIR score of **75**/**100**.

EDUCATION

University of Rajasthan **Jaipur, India**

Bachelor of Computer Applications (Honours with Research) | *CGPA*: **8**/**10** Oct 2023 – Jul 2027

Relevant Coursework: Artificial Intelligence and Machine Learning, Mathematics and Statistics, Data Science using Python, Data Structures & Algorithms (DSA), Data Communication, Cloud Computing.

Rajasthan Board of Secondary Education **Rajasthan, India**

Senior Secondary Education (Class XII): Aggregate*:* **81.80%** 2023

Secondary Education (Class X): Aggregate*:* **88.83%** 2021

HACKATHONS & COMPETITIONS

AI Mathematical Olympiad (AIMO) | *Kaggle* Kaggle

Solo Competitor Apr 2026

• Engineered *APEX-01*, integrating DeepSeek-R1 (**8B**), **SymPy** verification, multi-path sampling, and *NF4* mixed-precision inference under a **150**-second limit.

MIT Center for Quantum Engineering | *MIT iQuHack* GitHub

Participant Jan 2026

• Engineered hybrid *classical–quantum* algorithms for complex circuit modeling across NVIDIA, BlueQubit, hackathons, developing **gate-sequence** structural inference and *CPU*-based **state vector** simulations validated with *MPS* methods.

SKILLS

Research Skills

Physics-Informed Machine Learning & Deep Learning: Neural Operators, Koopman Operator Theory, Stochastic SINDy, Dynamic Mode Decomposition, ROM, Neural ODEs, *SE(3)*, EGNNs, Lie Groups.

Statistical Mechanics & Stochastic Dynamics: SDEs, Itô Calculus, Langevin Equation, Kramers–Moyal Expansion, Mori–Zwanzig Formalism, Fluctuation–Dissipation Theorem, Kalman Filtering.

Technical Skills

Programming Languages: Python, C++, Java, C#.

Scientific Frameworks: PyTorch, NumPy, Matplotlib, Scikit-Learn, PySINDy, PyDMD.

AI & Software: LLMs, RAG, MCP, Fine-Tuning, NLP, Docker, GitHub Actions, FastAPI, Node.js, MongoDB.

SEMINARS & WORKSHOPS

IBS BIMAG: *Reservoir Computing: Machine Learning Meets Nonlinear Dynamics* Oct 2026

ETH Zurich: *RDM Workshop 4: Reproducible Scientific Computing and Data Analysis* Mar 2026

MIT.nano: *Neuromorphic Computing via Integration of Diffusive and Drift Memristors* Dec 2025

ACHIEVEMENTS

Kaggle Expert: Ranked in the top **2**% globally, earned **4** bronze medals, and published **14** datasets. Aug 2026

LeetCode: Achieved global rank **81,544** and earned the *365*-Day Streak badge. May 2026

Stanford (*Code In Place*)**:** Completed **6**-week training in AI-assisted development and problem-solving. May 2026

ADDITIONAL ACTIVITIES

Aspire Institute: Participated in a **9**-week AI-driven leadership curriculum; earned **98**/**100**. Mar 2026

The Vision Lab, IISc Bangalore: *Cognitive Neuroscience Volunteer* Dec 2025